

```
In [ ]: import pandas as pd
import os
import time
import sys

import marimo as mo

from nilearn.glm.first_level import FirstLevelModel
from nilearn.interfaces.fmriprep import load_confounds
from nilearn.image import load_img
from nilearn.plotting import plot_design_matrix

import nilearn
print(f"nilearn version: {nilearn.__version__}")
```

nilearn version: 0.11.1

Set up directories and hyperparameters

```
In [ ]: # Hyperparameters
TR = 2.0
FWHM = 5.0
HIGH_PASS = 1/66

num_runs = 3
runs = [str(i + 1).zfill(2) for i in range(num_runs)]

current_dir = os.getcwd()
print(f"Current directory: {current_dir}")

# on HPC or on local machine?

# if HPC, get sub ID from command line args
if "dartfs" in current_dir:
    sub = sys.argv[1]
    sub = sub.zfill(2) # zero-pad to 2 digits if needed

    # sleep for 10s if not sub-01 - for creating save dirs on HPC
    if sub != "01":
        time.sleep(10)

# if local, manually set sub ID here
else:
    sub = "01"

print(f"Running sub-{sub}")
print(f"Hyperparameters: TR={TR}, FWHM={FWHM}, HIGH_PASS={HIGH_PASS:.6f}, ru
```

Current directory: /Users/gabe/Downloads/exam/univariateGLM
Running sub-01
Hyperparameters: TR=2.0, FWHM=5.0, HIGH_PASS=0.015152, runs=['01', '02', '03']

```
In [ ]: # Set directories

# on HPC or on local machine?
if "dartfs" in current_dir:
    data_dir = f"/dartfs/rc/lab/D/DBIC/DBIC/psych160/data/stopsignal/derivat
else:
    data_dir = f"../data/fmriprep/sub-{sub}/func"

save_dir = f"{current_dir}/output/sub-{sub}"
os.makedirs(save_dir, exist_ok=True)

report_save_dir = f"{current_dir}/output/reports"
os.makedirs(report_save_dir, exist_ok=True)
```

Load the data for this sub

```
In [ ]: fmri_imgs = []
events = []
confounds = []

for run in runs:

    if sub == "12" and run == "03":
        print(f"Skipping sub-{sub} run-{run} due to missing run.")
        continue

    events_df = pd.read_csv(f"{data_dir}/sub-{sub}_task-stopsignal_run-{run}")
    events_df = events_df.dropna(subset=['duration']) # remove 1st col, has
    events_df['trial_type'] = events_df['trial_type'].str.replace(' ', '_')
    events_df = events_df[["onset", "duration", "trial_type"]] # keep only r
    events.append(events_df)

    fmri_img_path = f"{data_dir}/sub-{sub}_task-stopsignal_run-{run}_space-M
    fmri_img = load_img(fmri_img_path)
    fmri_imgs.append(fmri_img)

    confound, _ = load_confounds(fmri_img_path,
                                strategy=["motion"],
                                motion="basic")

    confounds.append(confound)

assert len(events) == len(fmri_imgs) == len(confounds), f"Mismatch in events
print(f"Loaded {len(events)} runs of data for sub-{sub}.")
```

Loaded 3 runs of data for sub-01.

Fit the GLM

```
In [ ]: glm = FirstLevelModel(t_r=TR, high_pass=HIGH_PASS, smoothing_fwhm=FWHM, n_jobs=n_jobs)
        glm.fit(fmri_imgs, events=events, confounds=confounds)
```

▼ FirstLevelModel ⓘ

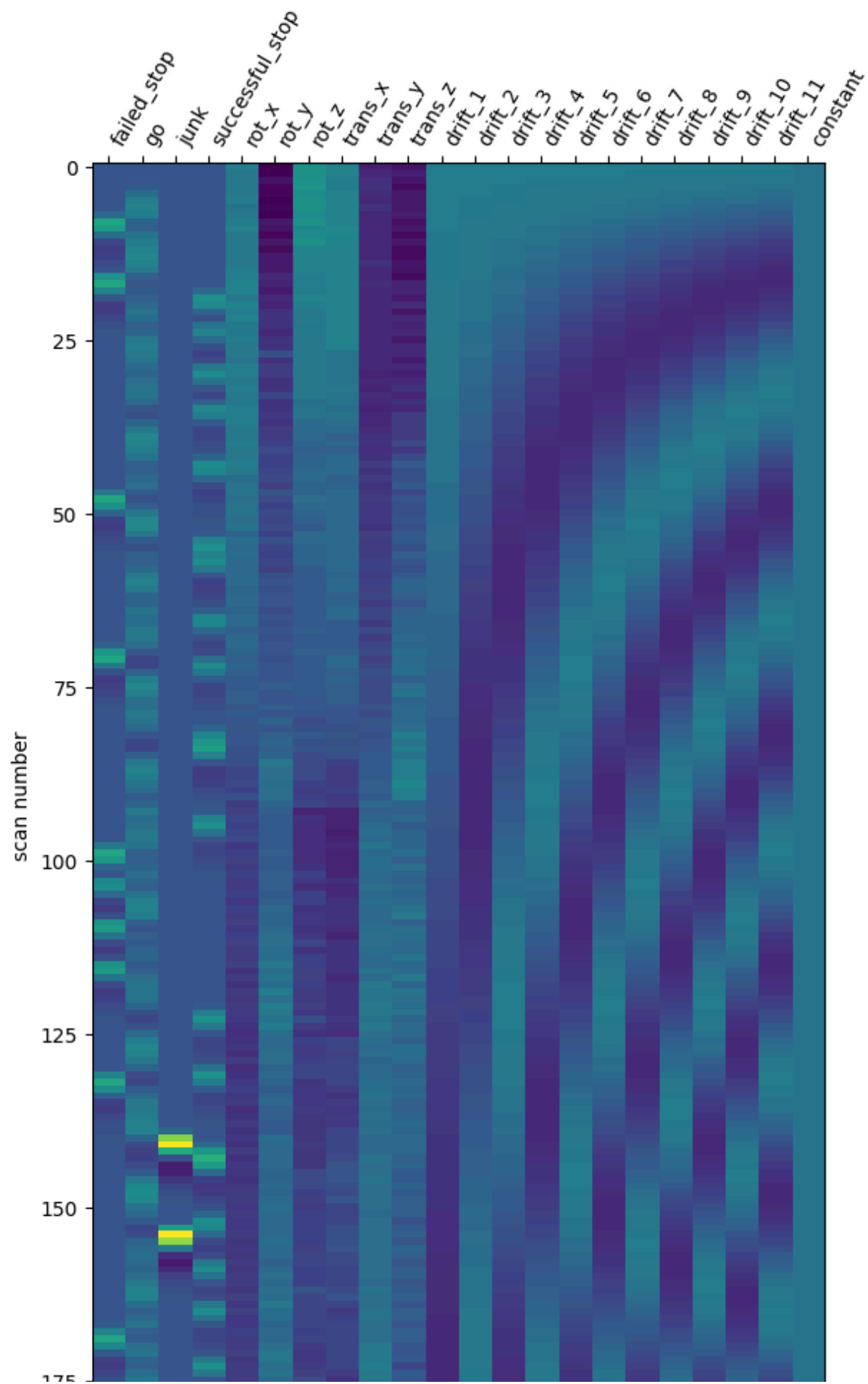
► Parameters

```
In [ ]: # compute desired contrasts and save
        contrasts = ["successful_stop - go"]

        for c in contrasts:
            beta_map = glm.compute_contrast(c, output_type="effect_size")
            beta_map.to_filename(f"{save_dir}/sub-{sub}_contrast-{c.replace(' ', '_')}")
        print(f"Computed contrasts and saved beta maps for sub-{sub} to {save_dir}.")
```

```
/var/folders/44/klw90pzs3fz633mk4sk_j1hc0000gn/T/marimo_4843/__marimo__cell_
BYtC_.py:5: UserWarning: One contrast given, assuming it for all 3 runs
    beta_map = glm.compute_contrast(c, output_type="effect_size")
Computed contrasts and saved beta maps for sub-01 to /Users/gabe/Downloads/ex
am/univariateGLM/output/sub-01.
```

```
In [ ]: # plot design matrix
        design_matrix = glm.design_matrices[0]
        plot_design_matrix(design_matrix)
```





```
In [ ]: # save GLM report
report = glm.generate_report(title=f"GLM Report for sub-{sub}", contrasts=cc
report.save_as_html(f"{report_save_dir}/sub-{sub}_glm_report.html")
print(f"Saved GLM report for sub-{sub} to {report_save_dir}.")
```

```
/Users/gabe/.cache/uv/archive-v0/CJu-ucKsZDiMVYTFM6iFi/lib/python3.11/site-p
ackages/nilearn/reporting/glm_reporter.py:545: UserWarning: One contrast giv
en, assuming it for all 3 runs
```

```
contrast_id: model.compute_contrast(
```

```
Saved GLM report for sub-01 to /Users/gabe/Downloads/exam/univariateGLM/outp
ut/reports.
```